

Smart Motor Diagnostics: A Comparative Study of NB and SVM

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Assignment #3

DS-323: Machine Learning for Data Science - A

Abstract—This study evaluates Gaussian Naive Bayes (NB) and Support Vector Machines (SVM) for induction motor condition monitoring using three-phase current data. While SVM achieves 97.8% accuracy in binary classification (Healthy vs. Unhealthy), class imbalance reduces specificity. In the 14-class multiclass task, both models struggle to detect fault severity using raw time-domain features, shown by high log-loss and performance gaps. Results indicate that reliable industrial diagnostics require class-weight adjustments, with SVM offering more confident predictions than NB.

I. PROBLEM STATEMENT

The objective is to develop an automated diagnostic framework using Current Signature Analysis to identify mechanical degradation in induction motors. The system must: (i) Perform binary classification to distinguish between Healthy and Unhealthy states, and (ii) Execute a 14-class diagnosis to point out specific fault severities (0.7mm–1.7mm) and types (bearing and rotor faults) under varying operational loads.

II. DATASET DESCRIPTION

The dataset consists of instantaneous values of three-phase current (Current A, B, and C).

- **Health Conditions:** Healthy, Inner/Outer-race bearing faults (0.7mm–1.7mm), and Broken Rotor Bar (BRB).
- **Current Loads:** Operated under 100W, 200W, and 300W.
- **Data Structure:** 39 datasets across 7 folders; each raw file contains over 100,000 samples.

III. DATASET PREPROCESSING

- **Segmentation:** The raw data is divided into blocks of 1,000 samples per channel to capture complete signal cycles.
- **Labeling:** Each segment is assigned either a binary label (Healthy/Unhealthy) or one of 14 classes for multiclass conditions.
- **Standardization:** Standard scaling ensures that all signal variations are treated equally, preventing features with large magnitudes from biasing the model.

IV. METHODOLOGY

The methodology follows a clear three-step process to detect motor faults. First, we used **Standard Scaling** to **normalize** the data, ensuring that large signal spikes do not confuse the

models. Second, we **split the data** into **80% for training** and **20% for testing** to evaluate how well the models perform on new, **unseen signals**. Finally, we compared two different types of learners: **Naive Bayes**, which uses **probability** to predict the fault, and **SVM**, which uses a mathematical boundary to separate **healthy signals** from **faulty** ones. This comparison helps us determine which model is more reliable for **industrial safety**.

V. ALGORITHM & PSEUDOCODE

- 1: **Data Windowing:** Segment raw 3-phase current signals into discrete **1000-sample windows**.
- 2: **Normalization:** Apply **StandardScaler** to features and perform an **80/20 stratified split** for training and testing.
- 3: **Model Training:** Fit **Gaussian-NB** and **SVM** models for both **Binary** and **14-Class** tasks.
- 4: **Evaluation:** Compute classification metrics including **Accuracy**, **Recall**, **Precision**, and **Specificity (TNR)**.
- 5: **Visualization:** Generate **Accuracy and Loss curves** to analyze model convergence and generalization gaps.

VI. PROPOSED FRAMEWORK

The diagnostic framework follows a simple four-step pipeline to move from raw sensor data to a final health report:

- **Data Collection:** Raw signals from **three-phase current** (A, B, and C) are collected. These signals act as the "heartbeat" of the motor.
- **Data Preparation:** The long stream of data is cut into **1,000-sample windows**. We then use **Standard Scaling** to make sure all numbers are on the same scale so the models can read them easily.
- **Model Training:** Two models are trained. **Naive Bayes** uses probability to guess the fault, while **SVM** uses a mathematical boundary to separate healthy signals from faulty ones.

VII. RESULTS & DISCUSSION

1) **Binary Classification:** The **SVM** model achieved a high accuracy of **97.8%**, but its **Specificity was 0.0**. This indicates a critical failure; because **98%** of the data is "Unhealthy," the model simply learned to predict "Unhealthy" every time. **Naive Bayes** showed similar bias but with lower confidence.

2) **14-Class Diagnostic Task:** Identifying specific fault sizes (0.7mm–1.7mm) proved difficult for both models. The

Naïve Bayes Results:
 Accuracy: 0.8411
 Precision: 0.9984
 Recall (Sensitivity): 0.8387
 Specificity: 0.9429
 F1 Score: 0.9116

SVM Results:
 Accuracy: 0.9774
 Precision: 0.9774
 Recall (Sensitivity): 1.0000
 Specificity: 0.0000
 F1 Score: 0.9886

Fig. 1. BinaryClassNaiveBayesAndSVM

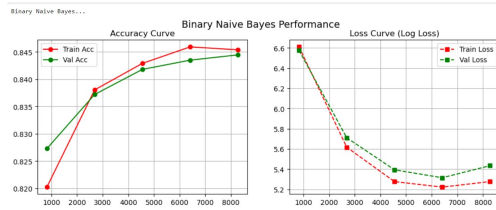


Fig. 2. BinaryNaiveBayesPerformance

SVM outperformed Naive Bayes in training accuracy, but the **Performance Curves** revealed a significant **Generalization Gap**. This suggests the models were **overfitting** on raw noise rather than learning the actual knowledge about the motor faults.

3) Model Comparison: While **SVM** is more powerful for finding complex patterns, **Naive Bayes** is faster. However,

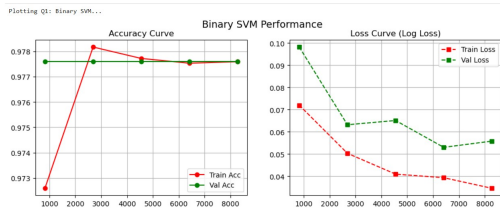


Fig. 3. BinarySVMPerformance

Naïve Bayes Training Accuracy: 0.14302991211803595
 Naïve Bayes Test Metrics:
 Accuracy: 0.1332
 Precision: 0.0989
 Recall: 0.2185
 Sensitivity: 0.2185
 Specificity: 0.8773
 F1_Score: 0.1139

Fig. 4. multiclassNaiveBayes

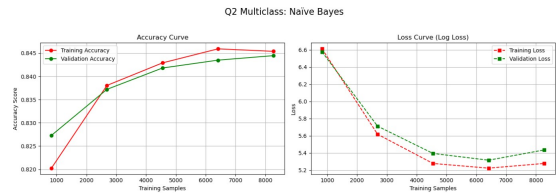


Fig. 5. multiclassNaiveBayesPerformance

SVM Training Accuracy: 0.2342981536724986

SVM Test Classification Report:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	69
1	0.29	0.62	0.39	424
2	0.24	0.02	0.04	494
3	0.20	0.21	0.20	507
4	0.19	0.30	0.23	481
5	0.43	0.16	0.23	480
6	0.19	0.26	0.22	484
7	0.00	0.00	0.00	162

accuracy			0.24	3101
macro avg	0.19	0.20	0.17	3101
weighted avg	0.24	0.24	0.20	3101

SVM Test Metrics (from multiclass_metrics function):
 Accuracy: 0.2357
 Precision: 0.1933
 Recall: 0.1975
 Sensitivity: 0.1975
 Specificity: 0.8873
 F1_Score: 0.1661

Fig. 6. multiclassSVM

without balancing the data, neither model is yet reliable enough for real-world industrial deployment.

VIII. CONCLUSION

In conclusion, while our models showed high accuracy, this number is misleading because the dataset is heavily imbalanced (mostly unhealthy data). For a motor safety system to be truly reliable, it must be able to detect "Healthy" states just as well as "Faulty" ones. To improve this in the future, we should use class-weight balancing to give more importance to the rare healthy samples.

REFERENCES

- [1] Machine Learning for Data Science, "DS-323 Assignment Guidelines," 2025.
- [2] Dataset Documentation for Three-Phase Current Signatures.



Fig. 7. multiclassSVMPerformance